Geometry Review: Unit 8 — Coordinate Geometry & Transformations

Answer Key

Station 1: Distance, Midpoint, and Slope in the Coordinate Plane

1. 5
2. 13
3. $5\sqrt{2} \approx 7.1$
4. $(1, 2)$
5. $B(5, 7)$
6. $-\dfrac{2}{3}$
7. 16; isosceles $AB = 6,\ AC = BC = 5$
8. Yes; $\overline{AB} \parallel \overline{DC}$ (slope $\frac{1}{3}$) and $\overline{AD} \parallel \overline{BC}$ (slope 3)
9. slope $PQ = \dfrac{1}{2}$, slope $QR = -2$; slopes are opposite reciprocals, so the angle at Q is a right angle
10. $(3, 4)$

Station 2: Equations of Circles

1. $x^2 + y^2 = 25$
2. $(x - 3)^2 + (y + 2)^2 = 16$
3. $(x + 1)^2 + (y - 5)^2 = 7$
4. center $(4, -1)$; $r = 6$
5. center $(0, 0)$; $r = 7$
6. center $(0, 3)$; $r = 2\sqrt{5}$
7. $(x - 2)^2 + (y - 3)^2 = 25$ $\quad r = \sqrt{3^2 + 4^2} = 5$
8. $(x-2)^2+(y-3)^2 = 20$ $\quad$ *center* $(2, 3)$*;* $r^2 = 4^2+2^2 = 20$
9. Yes, it is on the circle; $2^2 + 5^2 = 29 > 25$, so outside
10. $(x - 3)^2 + (y + 2)^2 = 25$; center $(3, -2)$, $r = 5$

Station 3: Translations, Reflections, and Rotations

1. $A'(6, -2)$
2. $(-2, 3),\ (0, 3),\ (-1, 6)$
3. $(4, 3)$
4. $(5, 2)$
5. $(7, 3)$
6. $(6, 5)$
7. $(-1, 3)$ $\quad (x, y) \to (-y, x)$
8. $(2, -5)$ $\quad (x, y) \to (-x, -y)$
9. $(-2, -4)$ $\quad (x, y) \to (y, -x)$
10. $(-1, -6)$ $\quad$ *same as* $90°$ *clockwise*

Station 4: Dilations and Compositions of Transformations

1. $(6, -9)$
2. $(2, 3)$
3. $A'(2, 4),\ B'(8, 4),\ C'(2, 12)$
4. 3
5. $(-4, 6)$
6. 20
7. $(5, 7)$ $\quad$ *image* $= (1 + 2 \cdot 2,\ 1 + 2 \cdot 3)$
8. $(5, -4)$
9. $(-3, 2)$ $\quad (2, 3) \to (-3, 2)$
10. 108 $\quad 12 \cdot 3^2$